

UNDERSTANDING THE BARRIERS TO PLANT-BASED DIETARY CHANGE: THE MEDIATING ROLE OF MEAT-CENTRIC MEAL PERCEPTION AMONG ADULTS IN AUSTRALIA AND DENMARK

Ramya Rao^{*1}, Dijn Gie Liem², and Qian Janice Wang¹

¹ Section for Design and Consumer Behaviour, Department of Food Science, University of Copenhagen

² Deakin Centre for Advanced Food Sciences, Deakin University

Email: s224972288@deakin.edu.au

INTRODUCTION

Shifting towards more plant-based diets can benefit both human and planetary health, yet reducing meat consumption remains challenging, particularly in meat-centric food cultures. Beyond preferences for individual foods, perceptions of what constitutes a “proper meal” may influence dietary choices. We investigated meat-centric meal perception - the tendency to perceive meat as a central component of a proper meal, and its association with barriers such as familiarity with plant-based meals, food neophobia, and cognitive reflection, and in turn its influence on plant-based meal consumption [1].

METHODS

An online questionnaire was completed by 592 meat-eating adults aged 20-70 years in Australia (n = 325) and Denmark (n = 267). Participants completed Likert-scale measures for four variables - familiarity with plant-based meals, food neophobia, cognitive reflection and meat-centric meal perception. Scales for familiarity with plant-based meals and meat-centric meal perception were designed based on ten interviews with Danish participants (age ranging 20-70). Participants also reported frequency of plant-based meal consumption. Mediation analyses within the Australian and Danish samples examined whether familiarity, food neophobia, and cognitive reflection were associated with meat-centric meal perception and if that turn was related to plant-based meal consumption. Age and sex were included as covariates.

RESULTS AND DISCUSSION

A stronger meat-centric meal perception was associated with consuming fewer plant-based meals in both Australia ($r = -0.36, p < .001$) and Denmark ($r = -0.42, p < .001$). Greater familiarity with plant-based meals was associated with a less meat-centric meal perception, which in turn was associated with greater plant-based meal consumption in both countries (Figure 1). Significant indirect effects were observed in Australia (0.34, 95% CI [0.183, 0.534]) and Denmark (0.45, 95% CI [0.267, 0.673]).

Food neophobia was associated with lower plant-based meal consumption in both samples. However, meat-centric meal perception mediated this relationship only in the Danish sample. Cognitive reflection was not significantly associated with meat-centric meal perception or plant-based meal consumption.

The findings suggest that barriers to plant-based eating extend beyond acceptance of individual foods to include broader ideas about what a meal should look like. Familiarity appears particularly important: people who were more familiar with plant-based meals were less likely to hold meat-centric meal perceptions and reported consuming more plant-based meals. The differences observed for food neophobia further suggest that the pathways underlying dietary change may depend on cultural context.

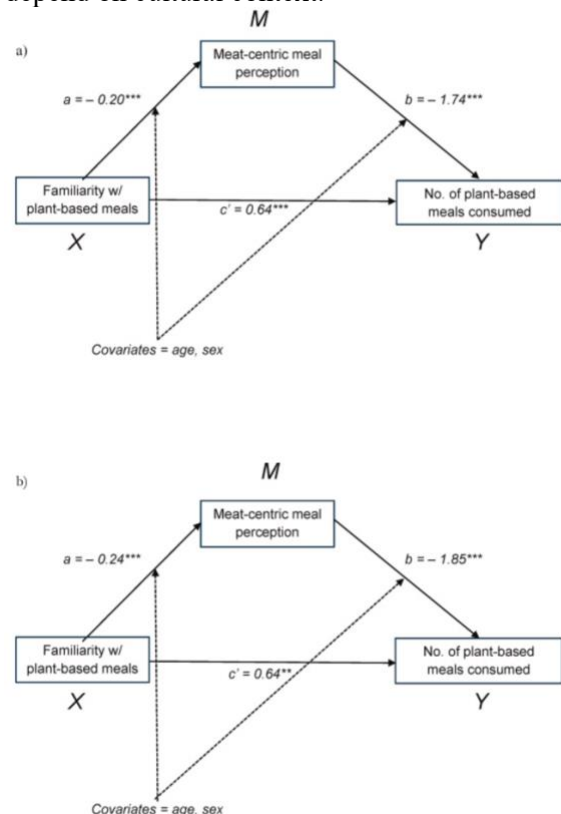


Figure 1a and 1b. Mediation models illustrating the relationship between familiarity with plant-based meals and the number of plant-based meals consumed, with meat-centric meal perception as the mediator. Figure 1a represents the Australian sample and 1b represents the Danish sample.

CONCLUSIONS

Viewing meat as central to a “proper meal” may represent an important psychological barrier to plant-based dietary change. Increasing familiarity with the structure, preparation and diversity of plant-based meals, rather than focusing solely on individual plant-based foods, may help challenge meat-centred meal norms and support more sustainable dietary choices.

REFERENCES

[1] Rao R et al. *Appetite* 215: 108240, 2025.

*Exchange student at Deakin University